Architecture for Galaxy on AnVIL in a VM

August 2026

Outline

An update on the *Galaxy on AnVIL v2* deployment model

Explorations for brining GalaxyAI to AnVIL

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What

Main goal: increase usability of the Galaxy service, and hence usage

How?

- Quicker startup
- Lower the costs
- Easier updates & bug fixes

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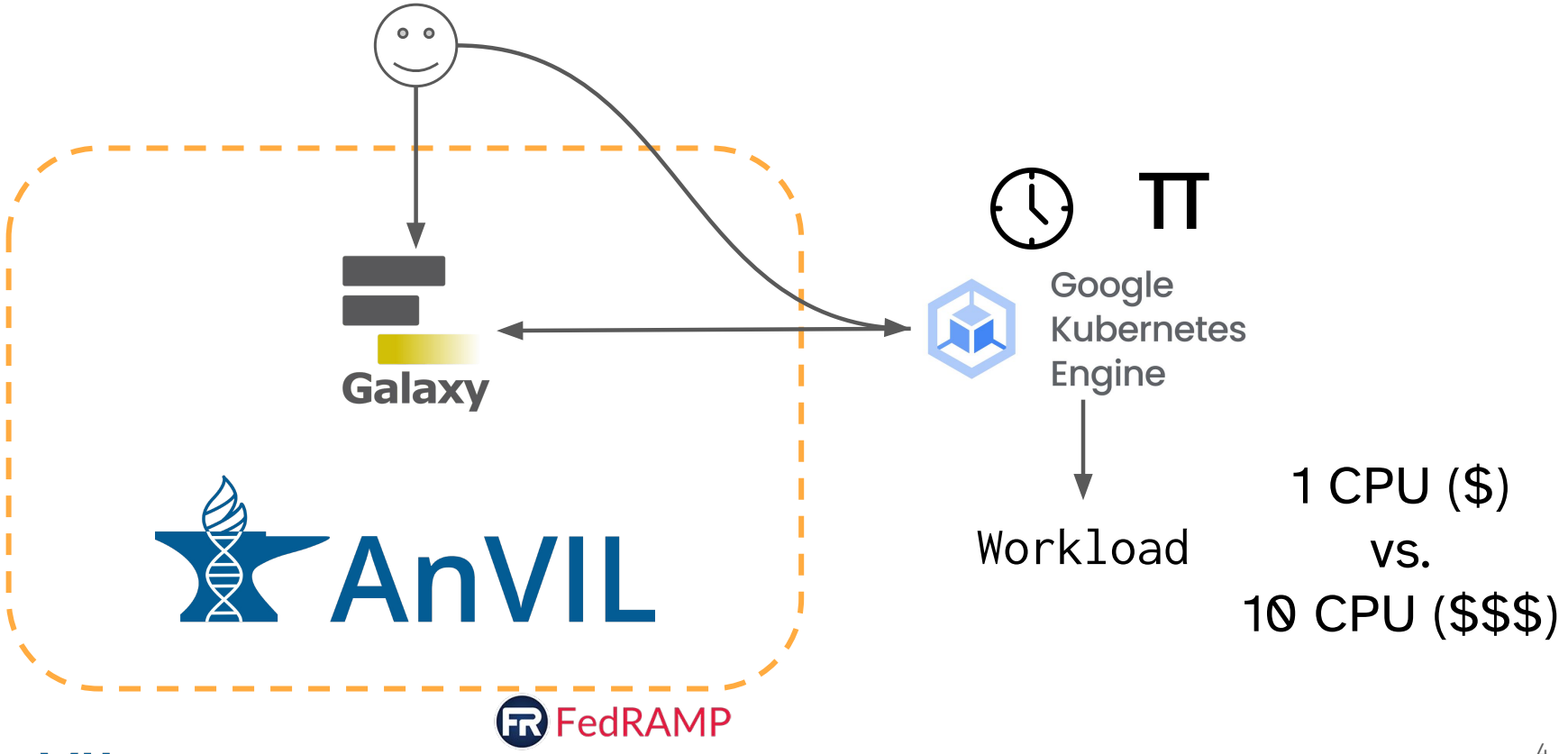
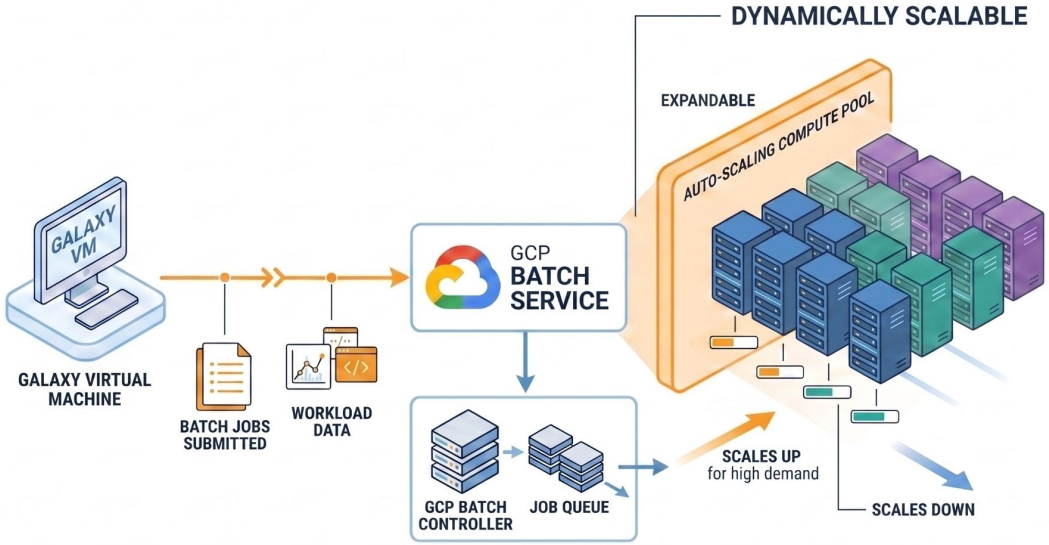
Galaxy on AnVIL today



Lightweight Galaxy VM, offloading jobs to GCP Batch for processing



Startup time

Before: ~15 min

Now: about the same 😔 *

Baseline compute cost

Now: $0.17/hr (~65%+ lower)

Before: $0.52+/hr

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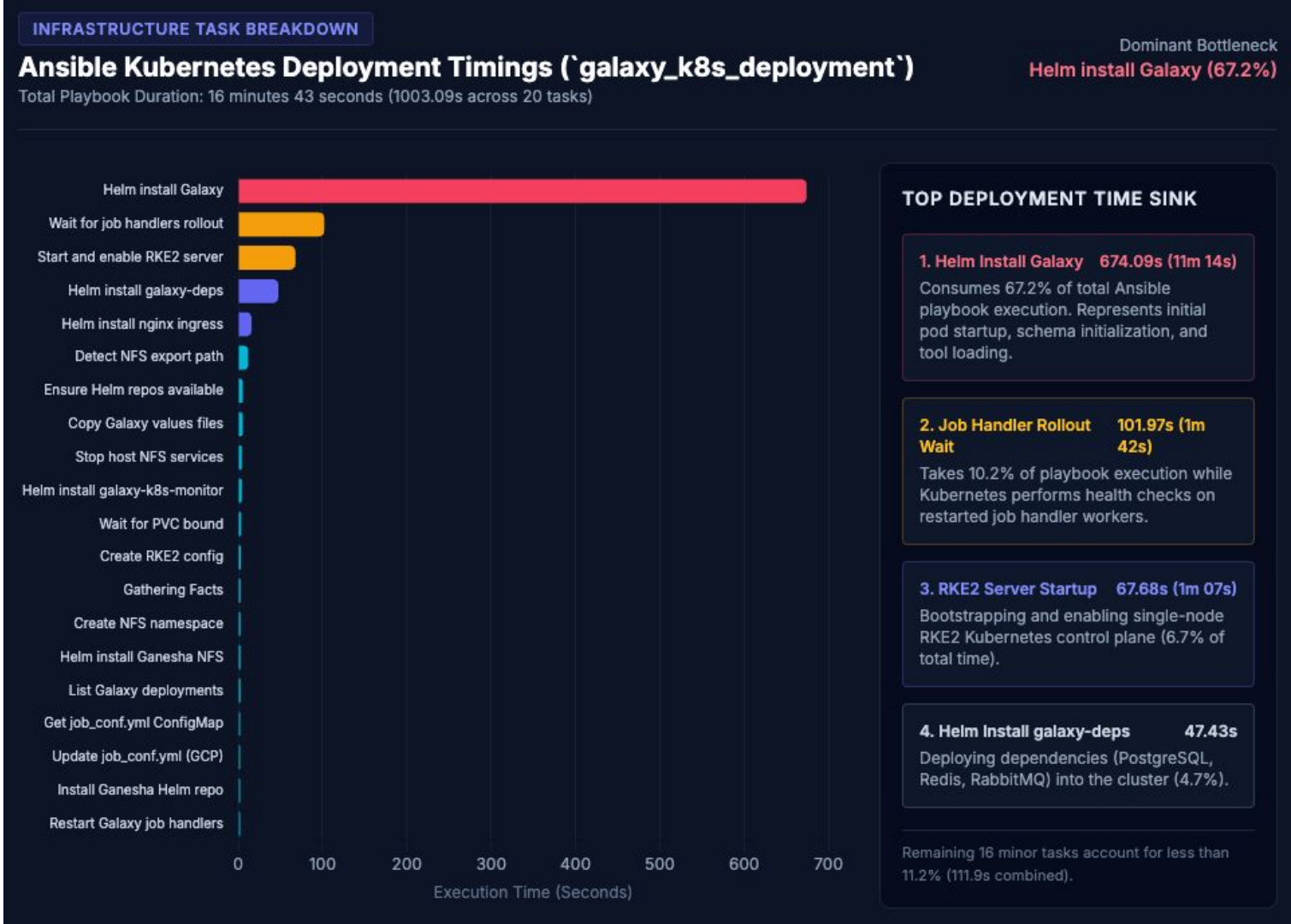
More on startup



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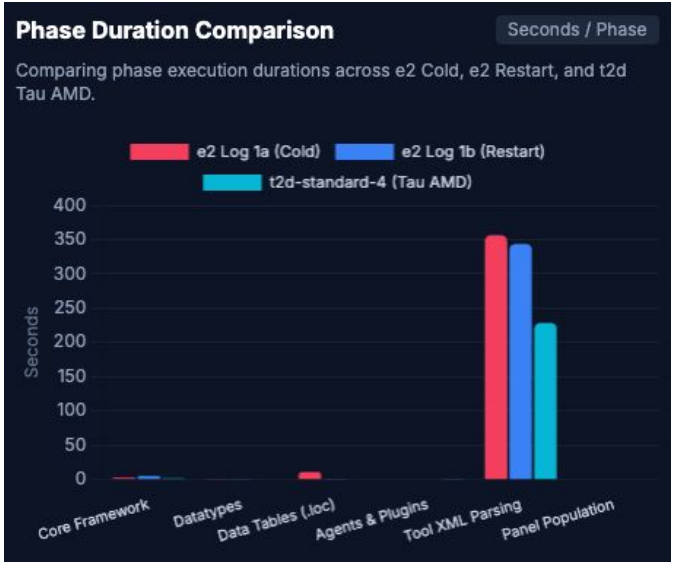
Timings for `helm install galaxy`



Reading hundreds files from CVMFS to build Galaxy's toolpanel is slow

> Galaxy 26.2 adds lazy toolbox loading, which should speed this up, hopefully to the target 5-6 mins

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Compute cost: proportional to the workload

Idle-time costs are substantially lower, and compound

Workload costs tend to be less than what a comparably-sized GKE cluster could process in the same amount of time

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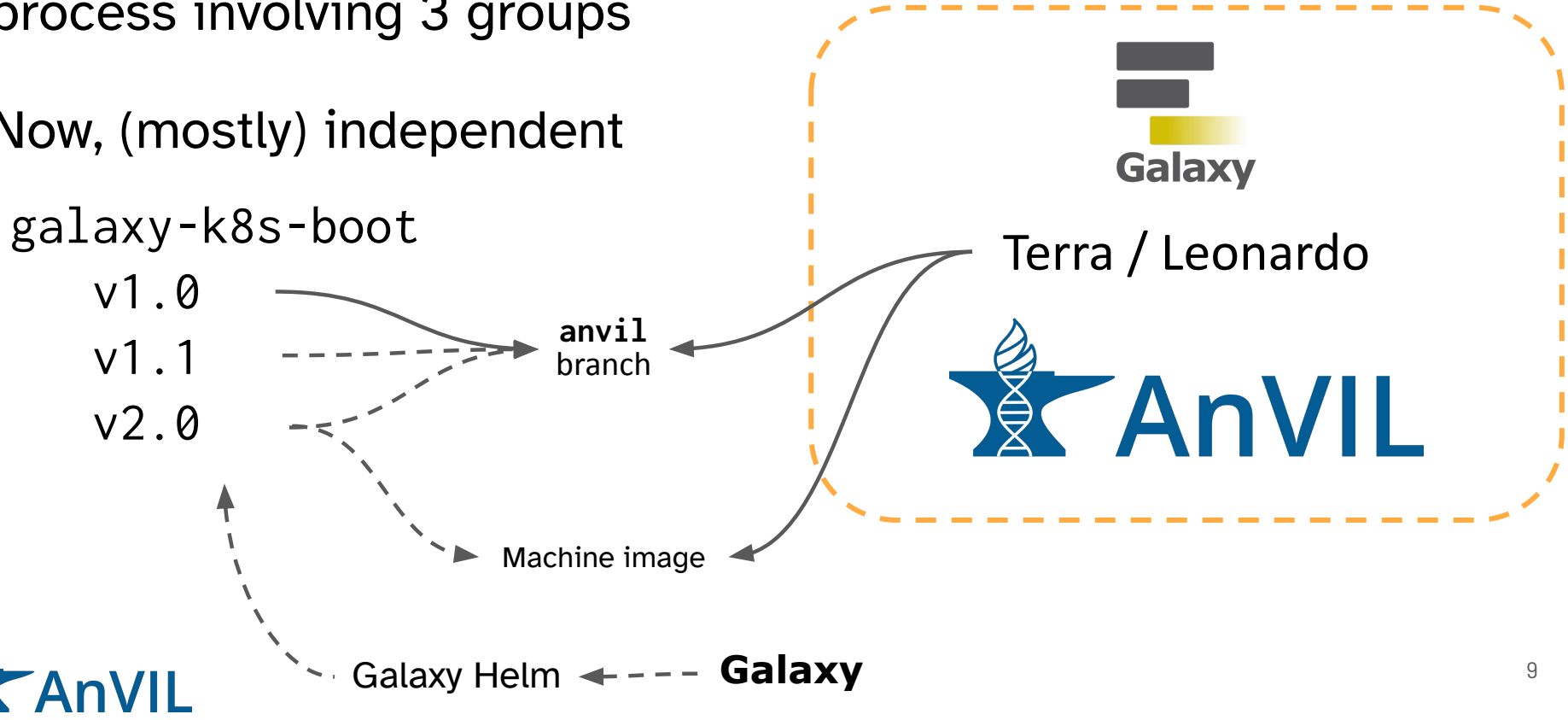
Largely independent maintenance

Upgrades were a month+ long process involving 3 groups

Now, (mostly) independent



Current status

Spent the past 2 weeks working on the integration (it's a slow, iterative process)

Core tested features check out!

Data import & export to Terra still has some issues.

Next steps: push to Terra Dev to allow broader testing, and then Prod. Target release announcement by ACC.

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Bringing AI to Galaxy on AnVIL

GalaxyAI and Orbit/Loom
Two paths to conversational analysis

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The elephant in the room

Is it allowed and/or feasible

For the purposes of this presentation we will assume, yes it is allowed.

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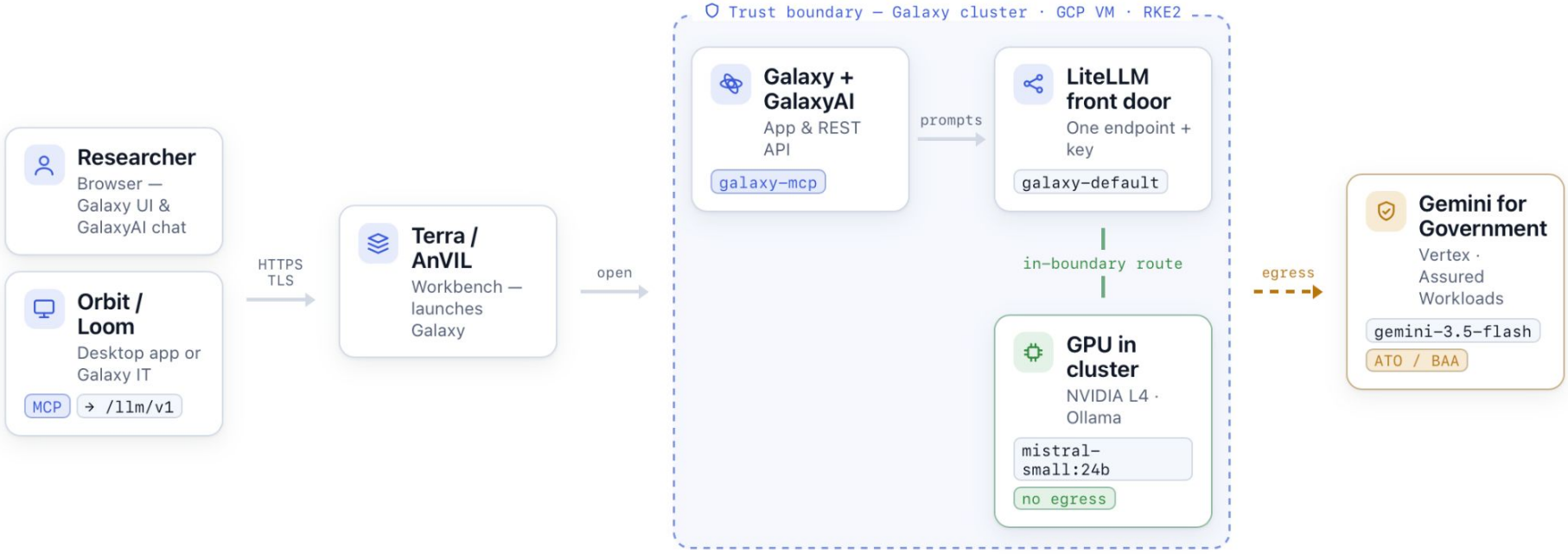
GalaxyAI Architecture



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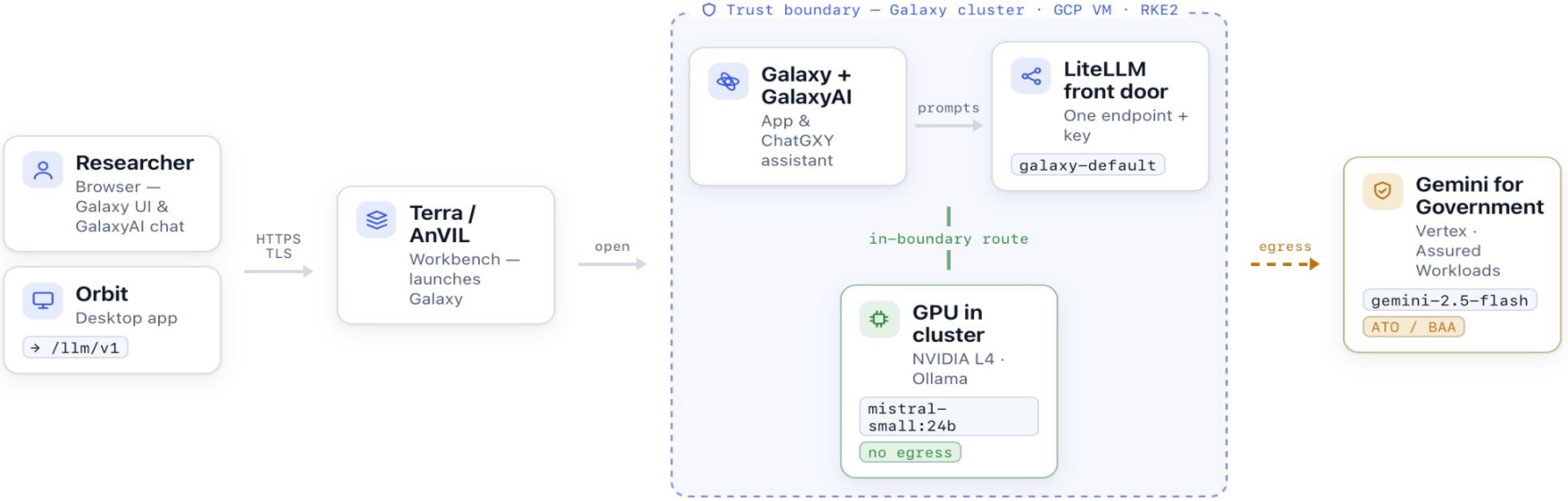
GalaxyAI — user access & LLM routing

galaxyai.useanvil.org · us-central1 · TLS



Trust boundary (cluster / VPC) · Self-hosted — data stays in-boundary · Egress — data leaves to a Google service

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Two complementary approaches

Natural-language help to discover tools, design workflows, troubleshoot analyses, and make sense of results

INSIDE GALAXY

GalaxyAI

An AI assistant built into Galaxy itself, rendered in the web UI and natively aware of your tools, workflows, and histories.

ALONGSIDE GALAXY

Orbit / Loom

A standalone AI agent client that connects to a Galaxy instance — on the desktop or launched inside Galaxy as an Interactive Tool.

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GalaxyAI

Server side, in the UI

- Part of the Galaxy application. No separate app to install

Natively Galaxy aware

- Agents understand the instance's tools, workflows, and histories directly.

Assists within Galaxy

- Find tools, design and repair workflows, and explain results in place.

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GalaxyAI

One front door, swappable back ends

- LiteLLM in-cluster
  Galaxy talks only to LiteLLM; the model behind it can change without touching Galaxy.
- No API key for Vertex (Google Gemini)
  LiteLLM authenticates with the VM service account (ADC over the metadata server).
- Deployed as config
  Turned on through *galaxy-k8s-boot*. Reproducible with the rest of the stack.

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Orbit / Loom

Loom is the framework

- An open agent runtime: a pi.dev based "brain," a web server, and MCP tool integration
- Communicates via Galaxy's REST API

Orbit is the client

- Runs standalone on the desktop
- Or as an Interactive Tool inside Galaxy

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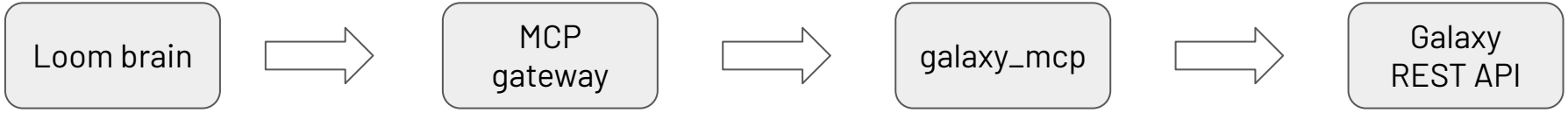
Orbit / Loom

How it reaches Galaxy



Speaks MCP

galaxy-mcp translates the agent's requests into Galaxy REST API calls.

Shared LLM path

The Loom *brain* points to the same LiteLLM front door

Acts as the user with the same permissions

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GalaxyAI vs Orbit / Loom

Dimension	GalaxyAI	Orbit / Loom
Where it runs	Server side, inside Galaxy	External client
How it reaches Galaxy	Internal API	REST API
Authentication	Instance wide	Per user
Scope	Galaxy only	General purpose agent
Model backend	One shared instance	Its own config

They can share the same LLM backend, but GalaxyAI is a *capability of Galaxy* while Orbit is a *client that talks to Galaxy*.

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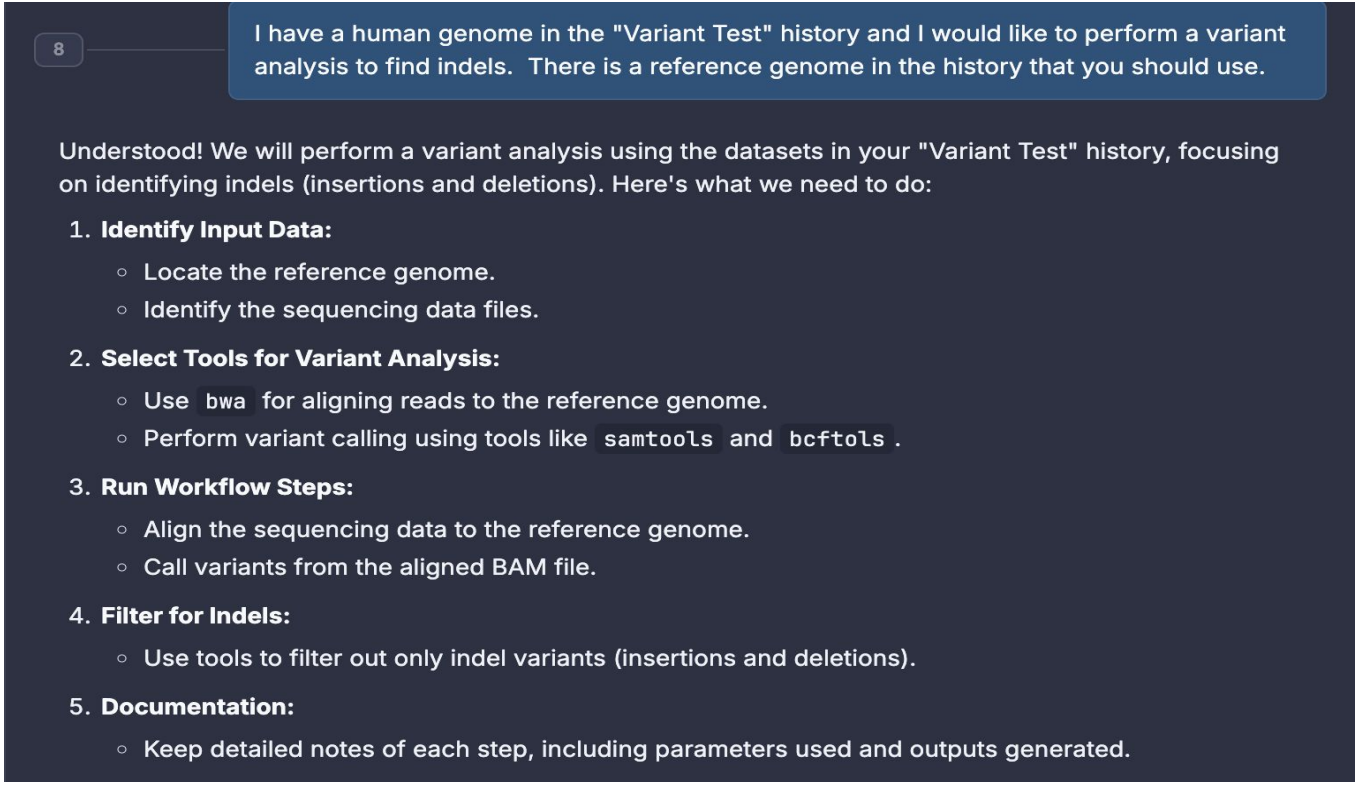
Examples - Orbit / Loom



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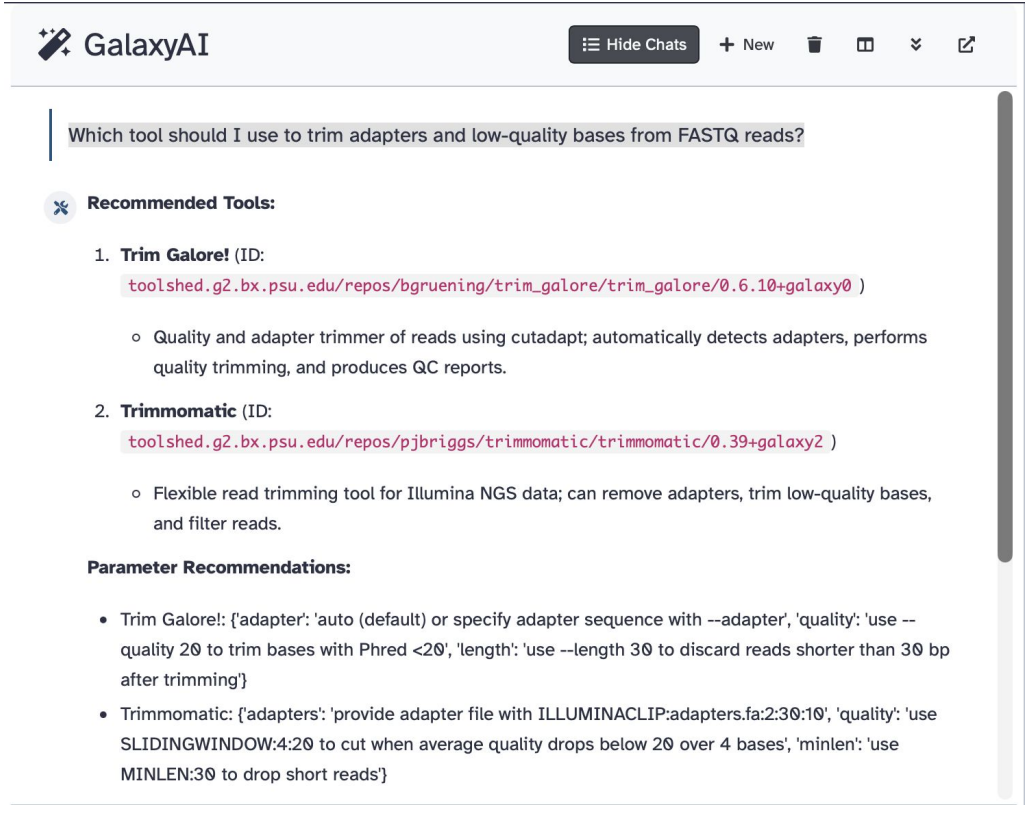
Examples - GalaxyAI



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Live Demo

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Questions

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